

## LEVEL 1 QUESTIONS

11. Print all **factors of a number**.

```
main.cpp  Run  Output  Clear
1 #include <iostream>
2 using namespace std;
3 int main() {
4     int n;
5     cout << "Enter a number: ";
6     cin >> n;
7     cout << "Factors are: ";
8     for (int i = 1; i <= n; i++) {
9         if (n % i == 0)
10            cout << i << " ";
11     }
```

Enter a number: 5  
Factors are: 1 5  
=== Code Execution Successful ===

12. Pyramid pattern.

```
main.cpp  Run  Output  Clear
1 #include <iostream>
2 using namespace std;
3 int main() {
4     int rows;
5     cout << "Enter number of rows: ";
6     cin >> rows;
7     for (int i = 1; i <= rows; i++) {
8         for (int j = 1; j <= rows - i; j++)
9             cout << " ";
10        for (int k = 1; k <= 2 * i - 1; k++)
11            cout << "**";
12        cout << endl;
13    }
14    return 0;
```

Enter number of rows: 5  
\*  
\*\*\*  
\*\*\*\*\*  
\*\*\*\*\*  
\*\*\*\*\*  
=== Code Execution Successful ===

### 13. Find the power of a number.

| main.cpp                                                                                                                                                                                                                                                                                                                                                                                 | Output                                                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| <pre>1 #include &lt;iostream&gt; 2 using namespace std; 3 int main() { 4     int base, power, result = 1; 5     cout &lt;&lt; "Enter base: "; 6     cin &gt;&gt; base; 7     cout &lt;&lt; "Enter power: "; 8     cin &gt;&gt; power; 9     for (int i = 1; i &lt;= power; i++) 10         result *= base; 11     cout &lt;&lt; "Result = " &lt;&lt; result; 12     return 0; 13 }</pre> | <pre>Enter base: 6 Enter power: 3 Result = 216  === Code Execution Successful ===</pre> |

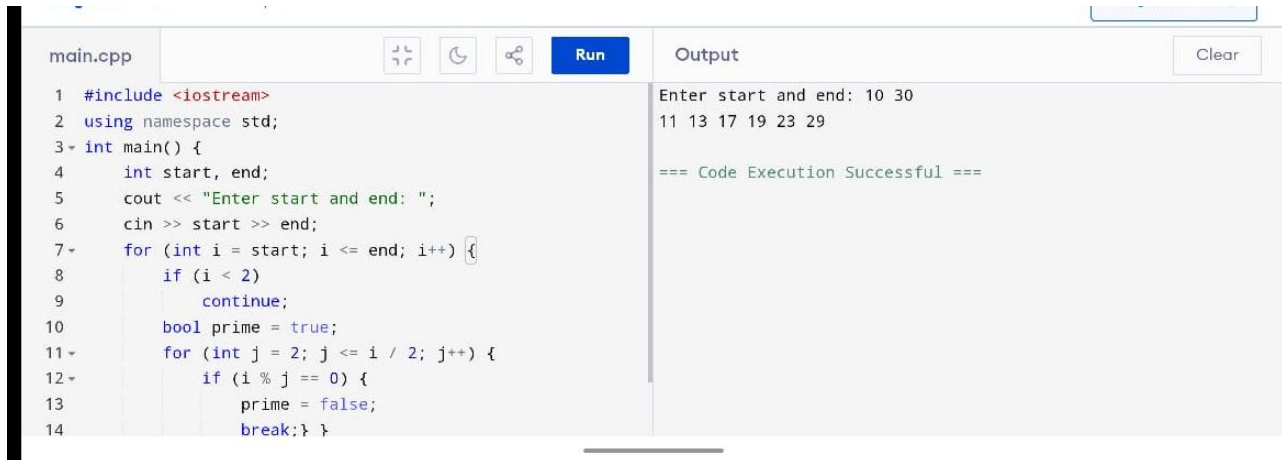
### 14. Inverted pyramid pattern.

| main.cpp                                                                                                                                                                                                                                                                                                                                                                                                                                       | Output                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| <pre>1 #include &lt;iostream&gt; 2 using namespace std; 3 int main() { 4     int rows; 5     cout &lt;&lt; "Enter number of rows: "; 6     cin &gt;&gt; rows; 7     for (int i = rows; i &gt;= 1; i--) { 8         for (int j = 1; j &lt;= rows - i; j++) 9             cout &lt;&lt; " "; 10        for (int k = 1; k &lt;= 2 * i - 1; k++) 11            cout &lt;&lt; "*"; 12        cout &lt;&lt; endl; 13    } 14    return 0; 15 }</pre> | <pre>Enter number of rows: 7 ***** ***** ***** ***** *** *  === Code Execution Successful ===</pre> |

### 15. Find the sum of positive and negative numbers from a list.

| main.cpp                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Output                                                                                                                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>1 #include &lt;iostream&gt; 2 using namespace std; 3 int main() { 4     int n, num, positive = 0, negative = 0; 5     cout &lt;&lt; "Enter number of elements: "; 6     cin &gt;&gt; n; 7     for (int i = 1; i &lt;= n; i++) { 8         cin &gt;&gt; num; 9         if (num &gt; 0) 10             positive += num; 11         else if (num &lt; 0) 12             negative += num; 13     } 14     cout &lt;&lt; "Sum of positive numbers = " &lt;&lt; positive &lt;&lt; endl; 15     cout &lt;&lt; "Sum of negative numbers = " &lt;&lt; negative; 16     return 0; 17 }</pre> | <pre>Enter number of elements: 4 10 60 7 23 Sum of positive numbers = 100 Sum of negative numbers = 0  === Code Execution Successful ===</pre> |

16. Print all **prime numbers** between two given numbers.



The screenshot shows a C++ IDE with a file named 'main.cpp'. The code is as follows:

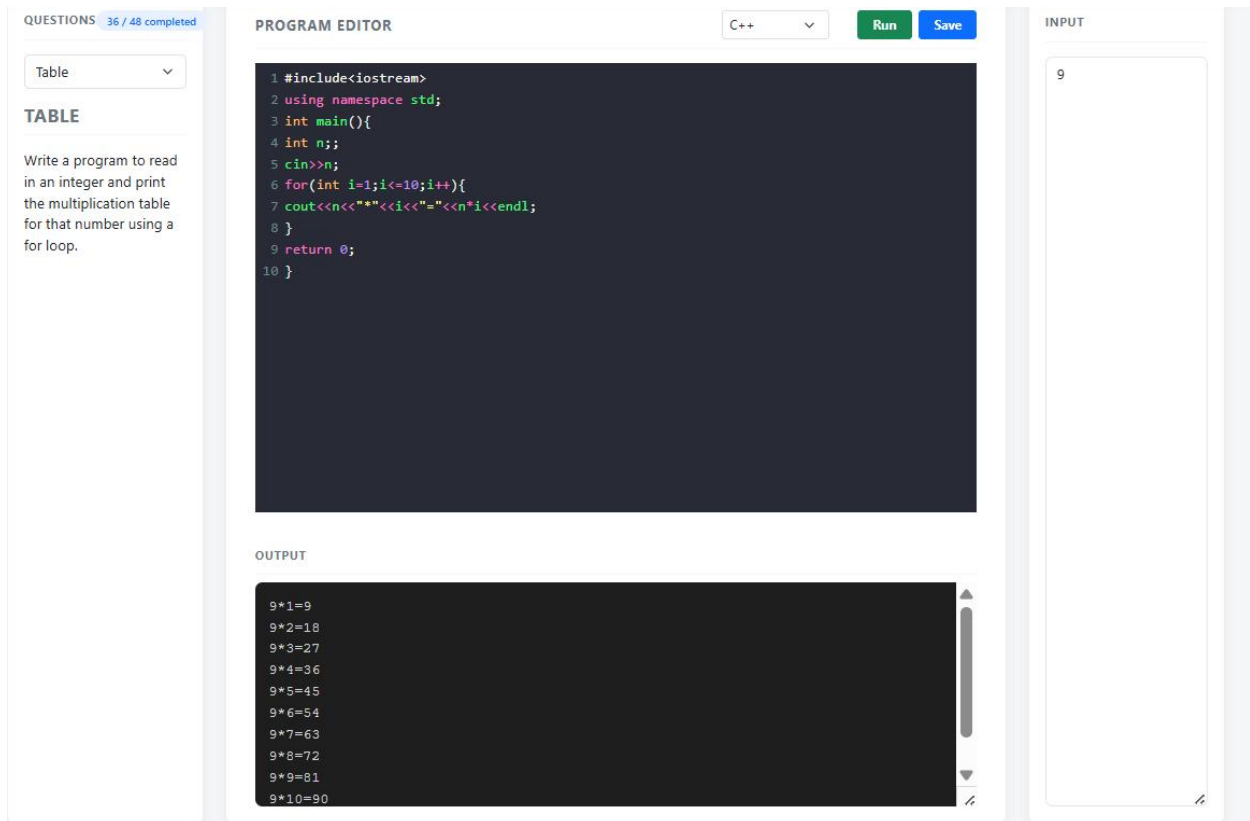
```
1 #include <iostream>
2 using namespace std;
3 int main() {
4     int start, end;
5     cout << "Enter start and end: ";
6     cin >> start >> end;
7     for (int i = start; i <= end; i++) {
8         if (i < 2)
9             continue;
10        bool prime = true;
11        for (int j = 2; j <= i / 2; j++) {
12            if (i % j == 0) {
13                prime = false;
14                break; } }
```

The output window shows the following text:

```
Enter start and end: 10 30
11 13 17 19 23 29

=== Code Execution Successful ===
```

17. Generate a **multiplication table**.



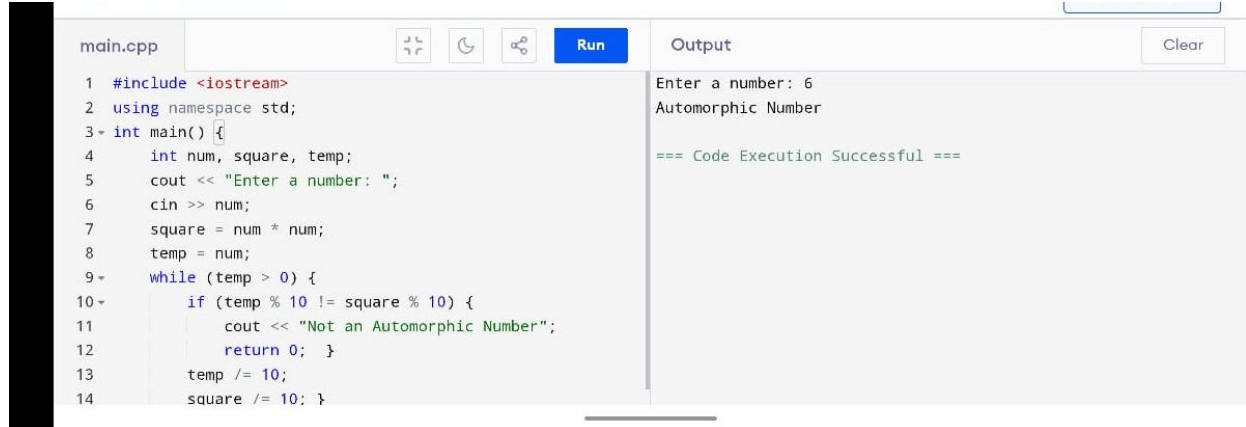
The screenshot shows a C++ IDE with a file named 'main.cpp'. The code is as follows:

```
1 #include<iostream>
2 using namespace std;
3 int main(){
4     int n;;
5     cin>n;
6     for(int i=1;i<=10;i++){
7         cout<<n<<"* "<<i<<"="<<n*i<<endl;
8     }
9     return 0;
10 }
```

The output window shows the following text:

```
9*1=9
9*2=18
9*3=27
9*4=36
9*5=45
9*6=54
9*7=63
9*8=72
9*9=81
9*10=90
```

18. Check whether a number is an **automorphic number**.



The screenshot shows a C++ IDE with a file named `main.cpp`. The code is as follows:

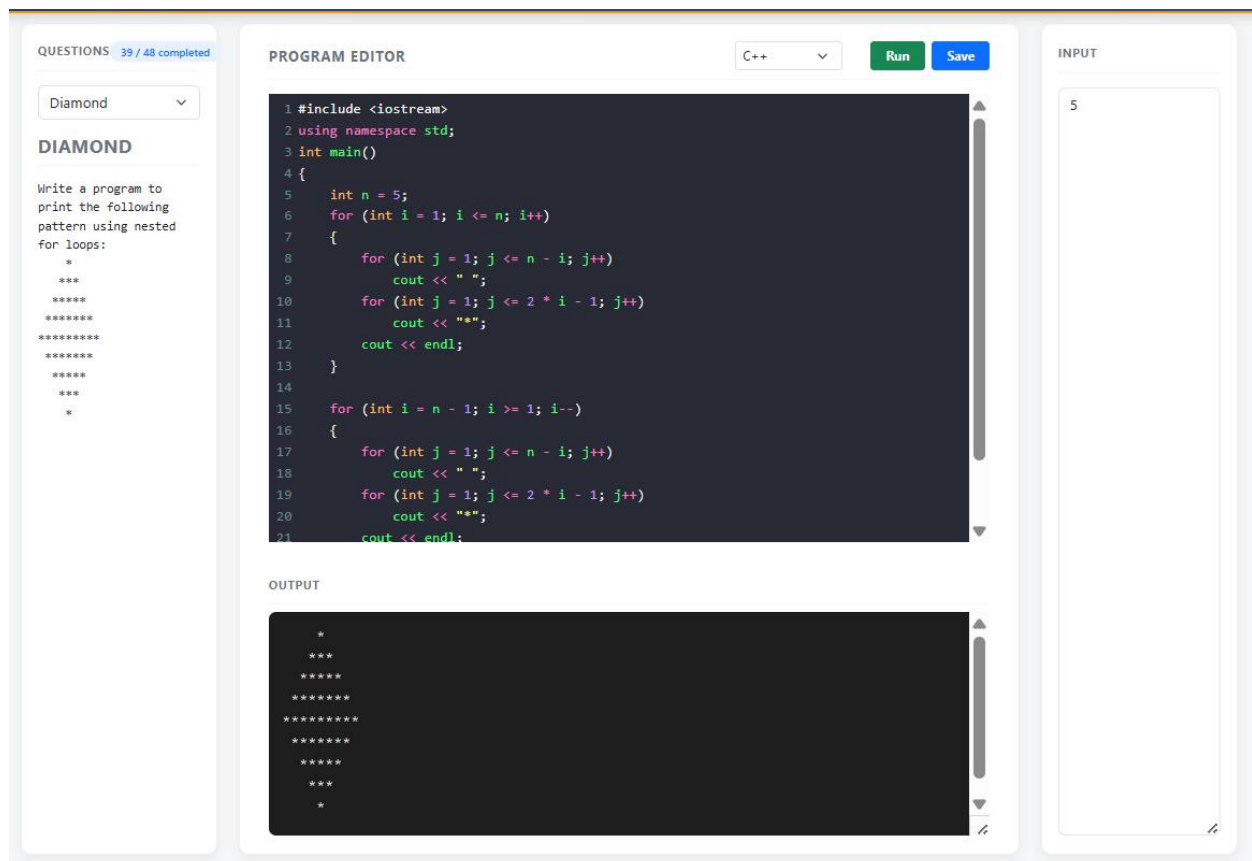
```
1 #include <iostream>
2 using namespace std;
3 int main() {
4     int num, square, temp;
5     cout << "Enter a number: ";
6     cin >> num;
7     square = num * num;
8     temp = num;
9     while (temp > 0) {
10        if (temp % 10 != square % 10) {
11            cout << "Not an Automorphic Number";
12            return 0; }
13        temp /= 10;
14        square /= 10; }
```

The output window on the right shows the following text:

```
Enter a number: 6
Automorphic Number

=== Code Execution Successful ===
```

19. Diamond pattern.



The screenshot shows a C++ IDE with a file named `main.cpp`. The code is as follows:

```
1 #include <iostream>
2 using namespace std;
3 int main()
4 {
5     int n = 5;
6     for (int i = 1; i <= n; i++)
7     {
8         for (int j = 1; j <= n - i; j++)
9             cout << " ";
10        for (int j = 1; j <= 2 * i - 1; j++)
11            cout << "*";
12        cout << endl;
13    }
14    for (int i = n - 1; i >= 1; i--)
15    {
16        for (int j = 1; j <= n - i; j++)
17            cout << " ";
18        for (int j = 1; j <= 2 * i - 1; j++)
19            cout << "*";
20        cout << endl;
21    }
```

The output window on the right shows the following text:

```

 *
 ***
 *****
 *******
 *****
 ***
 *
```

## 20.Convert hexadecimal to decimal and decimal to hexadecimal.

| main.cpp                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                        | Output |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|--------|
| <pre>1 #include &lt;iostream&gt; 2 #include &lt;iomanip&gt; 3 using namespace std; 4 int main() { 5     int decimal; 6     string hex; 7     cout &lt;&lt; "Enter decimal number: "; 8     cin &gt;&gt; decimal; 9     cout &lt;&lt; "Hexadecimal = " &lt;&lt; hex &lt;&lt; uppercase &lt;&lt;     decimal &lt;&lt; endl; 10    cout &lt;&lt; "Enter hexadecimal number: "; 11    cin &gt;&gt; hex; 12    int value; 13    cout &lt;&lt; dec;</pre> | <pre>Enter decimal number: 255 Hexadecimal = 255 Enter hexadecimal number: 745 Decimal = 1861  === Code Execution Successful ===</pre> |        |